

Now suppose another participant (participant A) enters the market with an order to buy Q shares over the same time and with the same urgency needs. This creates a buy market imbalance equal to the Q shares. Notice that this is equivalent to the Δq shares from our supply-demand above. The new buy shares are $V + Q$ and the sell shares remain at V . In order for these additional Q shares to execute buyers will have to provide a premium to the market to attract the additional sellers.

Let's define this total premium as I_S^* . Notice here that we are describing this process using dollar units. This process is the same whether dollars, dollars/share, or basis points are used. Our temporary impact parameter (b_1 from Equation 4.7) defines the breakdown between temporary and permanent cost. Total temporary cost is $b_1 \cdot I_S^*$ and total permanent cost is $(1 - b_1) \cdot I_S^*$.

In this formulation, it is not fair to assume that the entire temporary cost will be borne by participant A alone. The temporary cost will be rather shared (allocated) across all buyers. Think of this approach as an average costing methodology.

Since we now expect there to be $V + Q$ shares traded, that is, the original V shares plus the newly arrived Q shares, the portion of total temporary impact expected to be borne by investor A is calculated in proportion to their total trade volume. This is:

Cost Allocation Method

Temporary market impact cost is dependent upon the underlying trading rate. This rate is expressed in terms of percentage of volume or simply POV. It is:

$$POV = \frac{Q}{Q + V}$$

In this notation, Q is the net imbalance (absolute difference between buying and selling pressure), V is the expected volume excluding the order imbalance, and $Q + V$ is the total number of shares that is expected to trade in the market.

Therefore we have,

$$\text{Temporary Impact} = b_1 \cdot I_S^* \cdot \frac{Q}{Q + V} \quad (4.8)$$

$$\text{Permanent Impact} = (1 - b_1) \cdot I_S^* \quad (4.9)$$